Detailed BTP Progress Report

Resilient Swarm Navigation via Trust-Aware Multi-Agent Reinforcement Learning (TA-MAPPO)

Submitted By

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1 Abstract

Autonomous drone swarms are increasingly used in applications such as surveillance, search-and-rescue operations, autonomous logistics, and military reconnaissance. These systems require reliable decentralized coordination among multiple agents operating simultaneously in dynamic environments. However, traditional Multi-Agent Reinforcement Learning (MARL) frameworks assume honest communication between all participating agents. In real-world deployments, compromised drones may intentionally broadcast deceptive information, leading to swarm instability, collisions, and mission failure.

This project proposes a novel Trust-Aware Multi-Agent Proximal Policy Optimization (TA-MAPPO) framework for resilient decentralized swarm navigation under adversarial communication conditions. The framework combines bio-inspired trust evaluation mechanisms with MARL to suppress unreliable communication before policy inference. The system is implemented using a continuous-control PettingZoo-based drone swarm simulator integrated with Stable-Baselines3 PPO training.

Experimental results obtained during Phase A demonstrate highly stable swarm convergence with a success rate of 99.68% in random-spawn scenarios and 95.78% in dense-cluster environments. The project establishes a strong foundation for future adversarial resilience research in decentralized swarm intelligence.

2 Brief Project Overview

2.1 Problem Statement

Autonomous drone swarms are becoming increasingly important in robotics applications such as surveillance, disaster response, military reconnaissance, and warehouse automation. Swarm intelligence enables multiple drones to coordinate collectively and solve tasks efficiently in decentralized environments.

Despite recent progress in Multi-Agent Reinforcement Learning, existing frameworks assume reliable communication between all agents. This assumption becomes unrealistic in adversarial environments where compromised drones may intentionally transmit deceptive positional information or corrupted sensor data.

Such adversarial behavior can create:

- Swarm fragmentation
- Collision cascades
- Navigation instability
- Congestion deadlocks
- Mission failure

Traditional MARL systems lack mechanisms to proactively evaluate the trustworthiness of incoming communication. Most methods react only after failures occur, which is unsafe in critical swarm systems.

Therefore, there is a need for a trust-aware resilient swarm navigation framework capable of detecting unreliable communication while preserving decentralized coordination.

2.2 Objectives

The major objectives of this project are:

1. Design a trust-aware decentralized swarm coordination framework.
2. Develop a continuous-space multi-agent reinforcement learning environment.
3. Integrate trusted LiDAR sensing with untrusted communication channels.
4. Implement bio-inspired trust evaluation for adversarial detection.
5. Improve collision avoidance and congestion handling in dense swarm environments.
6. Build a scalable and reproducible swarm evaluation framework.
7. Extend traditional MAPPO into a robust TA-MAPPO architecture.

2.3 Research Motivation

Drone swarms are increasingly deployed in environments where communication integrity cannot always be guaranteed. Existing reinforcement learning frameworks prioritize coordination efficiency but often ignore adversarial robustness.

Biological immune systems provide a natural model for anomaly detection and selective trust regulation. T-cell immune mechanisms continuously evaluate signals and suppress harmful interactions before they affect the overall system.

Inspired by these principles, this project integrates bio-inspired trust filtering within a swarm reinforcement learning framework. The research combines:

- Bio-inspired anomaly detection
- Trust-aware communication
- Decentralized swarm intelligence
- Multi-Agent Reinforcement Learning

No existing framework currently integrates trust gating, CTDE learning, continuous-control swarm navigation, and adversarial resilience within a unified drone swarm environment.

2.4 Expected Contribution

The expected contributions of this project include:

2.4.1 TA-MAPPO Framework

A trust-aware extension of MAPPO integrating bio-inspired trust evaluation into decentralized swarm reinforcement learning.

2.4.2 Trust-Gated Communication

A proactive trust-filtering system that suppresses deceptive communication before policy inference instead of penalizing malicious agents after failure occurs.

2.4.3 Social-Distancing Reward

A novel reward-engineering mechanism designed to reduce collision cascades and improve dense swarm coordination.

2.4.4 Reproducible Swarm Environment

A PettingZoo-based continuous multi-agent simulator supporting:

- Obstacle navigation
- Dense-cluster testing
- Adversarial communication

- Statistical reproducibility

3 Methodology and System Design

3.1 Proposed Approach

The proposed framework uses Trust-Aware Multi-Agent Proximal Policy Optimization (TA-MAPPO) for decentralized drone swarm coordination.

The environment consists of:

- 10 autonomous drones
- Continuous 20×20 navigation space
- Shared navigation targets
- LiDAR-based sensing
- Inter-drone communication

The framework follows the Centralized Training with Decentralized Execution (CTDE) paradigm. During training, the critic receives global swarm information while the actor receives only local observations.

3.2 Trust-Aware Communication Framework

Each drone receives:

1. Trusted LiDAR observations
2. Untrusted communication vectors from neighboring drones

A bio-inspired trust evaluator computes trust scores:

$$T_{ij} = \sigma \left(\sum_{k} w_k \left(o_{ik}^{trusted} - b_{jk}^{broadcast} \right) \right)$$

where:

- T_{ij} = trust score assigned to neighbor j
- $o_{ik}^{trusted}$ = trusted LiDAR observation
- $b_{jk}^{broadcast}$ = received communication

Low-trust communication vectors are attenuated before policy inference.

3.3 Algorithm Workflow

The TA-MAPPO workflow consists of:

1. Observation collection
2. Trust evaluation
3. Feature gating
4. PPO policy inference
5. Continuous action execution
6. Reward optimization

Each drone collects local sensor information and communication vectors. The trust module evaluates consistency between trusted sensor information and received broadcasts. Low-trust features are suppressed before policy inference. The actor network outputs continuous velocity commands (v_x, v_y) which are executed inside the environment.

3.4 CTDE Architecture

3.4.1 Centralized Critic

The critic receives the full global swarm state during training. Each drone contributes:

- Position
- Velocity
- LiDAR sweep

For a 10-drone swarm:

$$10 \times 52 = 520$$

global critic features are used.

3.4.2 Decentralized Actor

The actor uses only local observations:

- Self-kinematics
- LiDAR readings
- Neighbor communication
- Temporal context
- Trust anomaly indicators

The actor observation space contains approximately 130 features.

3.5 Simulation Framework

The simulation environment is implemented using:

- PettingZoo ParallelEnv API
- Stable-Baselines3 PPO
- PyTorch neural networks
- NumPy
- Matplotlib
- PyGame rendering engine

The environment supports continuous drone dynamics, LiDAR sensing, collision handling, obstacle navigation, and dense swarm evaluation.

4 Implementation Progress

4.1 Modules Developed

4.1.1 Swarm Environment

A complete continuous-space drone swarm environment has been implemented with:

- Continuous velocity control
- Physics-based movement
- Collision handling
- Goal navigation
- Multi-agent synchronization

4.1.2 LiDAR System

The environment currently supports:

- 16-ray LiDAR sensing
- Wall detection
- Drone proximity sensing
- Sensor normalization

4.1.3 Reward Engineering

Several reward functions have been implemented:

- Goal-directed reward
- Cohesion reward
- Cluster-expansion reward
- Social-distancing penalty
- Speed-limit regulation
- Collision penalties

4.1.4 Evaluation Framework

The evaluation pipeline supports:

- Random scenario generation
- Clustered stress testing
- Edge-case testing
- Statistical reproducibility

4.2 Code Architecture

The codebase is modularized into:

- Environment layer
- PPO training layer
- Evaluation layer
- Visualization layer

This modular design improves maintainability and future adversarial extensions.

4.3 Integration Status

Phase	Status
Phase A - Basic Swarm Coordination	Completed
Phase B - Obstacle Navigation	In Progress
Phase C - Trust Evaluation	Pending
Phase D - Full Adversarial Defense	Pending

Table 1: Current Integration Status

4.4 Current Completion Level

The project is currently estimated to be approximately 60% complete.

Completed components:

- Environment implementation
- PPO integration
- Reward engineering
- Evaluation framework

Pending components:

- Trust gating integration
- Adversarial communication
- Full TA-MAPPO evaluation

5 Simulation Setup

5.1 Simulation Environment

Parameter	Value
Arena Size	20×20
Number of Drones	10
Time Step	0.1 s
Maximum Episode Length	600 steps
Maximum Velocity	2.0
LiDAR Rays	16
Action Space	Continuous (v_x, v_y)

Table 2: Simulation Parameters

5.2 Training Parameters

Hyperparameter	Value
RL Algorithm	PPO
Learning Rate	5×10^{-5}
Entropy Coefficient	$0.005 - 0.03$
Rollout Workers	10
Reward Normalization	Enabled
Observation Normalization	Disabled

Table 3: Training Hyperparameters

5.3 Test Scenarios

The environment supports:

- Random spawn scenarios
- Dense cluster configurations
- Obstacle navigation maps
- Future adversarial communication settings

5.4 Performance Metrics

The following metrics are used:

- Success Rate
- Collision Rate
- Time to Goal
- Convergence Stability

6 Results Obtained So Far

6.1 Preliminary Simulation Results

Scenario	Success Rate	Standard Deviation
Random Spawn	99.68%	±0.19
Dense Cluster	95.78%	±0.42

Table 4: Phase A Evaluation Results

6.2 Comparative Analysis

Traditional Vanilla MAPPO systems suffer from:

- Dense-cluster instability
- Swarm panic behavior
- Collision cascades

The proposed framework significantly improves coordination performance in dense swarm conditions. The dense-cluster success rate improved from:

$$89.45\% \rightarrow 95.78\%$$

6.3 Interpretation of Results

The obtained results demonstrate:

- Stable decentralized coordination
- Improved congestion handling
- Effective collision avoidance
- Reliable continuous-control navigation

These findings validate the feasibility of extending the framework toward adversarial resilience.

7 Novelty and Research Contribution

7.1 Novelty of the Proposed Work

The novelty of this work lies in integrating:

- Bio-inspired trust evaluation
- Trust-aware communication filtering
- Multi-Agent Reinforcement Learning
- Continuous decentralized swarm control

within a unified framework.

7.2 Improvements Over Existing Methods

Existing MARL systems:

- Assume reliable communication
- React only after failures occur
- Lack proactive trust evaluation

The proposed TA-MAPPO framework suppresses deceptive information before policy inference and improves dense-swarm resilience.

7.3 Contribution to the Field

Potential contributions include:

- Byzantine-resilient swarm intelligence
- Adversarially robust MARL systems
- Bio-inspired trust-aware robotics
- Reproducible adversarial swarm simulators

8 Publication Readiness

8.1 Target Journals and Conferences

Potential publication venues include:

- IEEE Transactions on Intelligent Transportation Systems
- Engineering Applications of Artificial Intelligence
- IEEE ICRA Workshops
- AAMAS MARL Workshops

8.2 Tentative Paper Title

TA-MAPPO: Trust-Aware Multi-Agent Reinforcement Learning for Resilient Drone Swarms

8.3 Proposed Paper Outline

1. Introduction
2. Related Work
3. TA-MAPPO Framework
4. Trust-Gated Architecture
5. Experimental Setup
6. Results and Analysis
7. Ablation Studies
8. Conclusion and Future Work

8.4 Additional Experiments Required

Before publication-quality submission, the following experiments are required:

- Full adversarial Phase C implementation
- Trust-score convergence analysis
- ROC curve evaluation
- Confusion matrix analysis
- Baseline comparisons
- Scalability testing

9 Challenges and Future Work

9.1 Current Limitations

Current limitations include:

- No active traitor implementation

- Only 2D simulation support
- Static obstacle assumptions
- Limited scalability evaluation

9.2 Technical Challenges Faced

Major technical challenges include:

- Dense swarm stability
- Reward balancing
- PPO convergence stability
- Environment solvability

9.3 Planned Improvements

9.3.1 Near-Term Improvements

- Implement deceptive communication
- Add trust-score heatmaps
- Improve obstacle navigation

9.3.2 Mid-Term Improvements

- Full TA-MAPPO adversarial evaluation
- Baseline comparisons
- Hyperparameter optimization

9.3.3 Long-Term Improvements

- Transition to realistic 3D environments
- PX4 integration
- Hardware-in-the-loop testing

9.4 Projected Timeline

Timeline	Planned Work
Next 2 Weeks	Complete Phase B
Next 1 Month	Implement Phase C
Next 2 Months	Full TA-MAPPO Evaluation
Final Stage	Publication Experiments

Table 5: Projected Timeline

10 Conclusion

This project establishes a strong foundation toward resilient decentralized swarm navigation under adversarial communication conditions. The completed implementation demonstrates highly stable continuous-control swarm coordination and effective congestion handling in dense multi-agent environments.

The proposed TA-MAPPO framework combines:

- Multi-Agent Reinforcement Learning
- Bio-inspired trust mechanisms
- Trust-aware communication filtering
- Continuous decentralized control

Future work will focus on adversarial communication modeling, trust-aware defensive evaluation, large-scale swarm testing, and realistic hardware deployment.